

Final Consolidated Technical Solution Plan: Archive Intelligence Assistant

Executive Summary

The proposed solution is a **provenance-first, multimodal archive intelligence assistant** for the Ashen Era Archive. It selects **Sub-track 1C — Searching the Way a Human Does** as the primary track and implements **Sub-track 1B — Connecting Facts Across Thousands of Pages** as a core supporting capability. Selected Sub-track 1A behavior is also included through table, figure, diagram, map, and image retrieval.

The system reads every archive format, preserves page- and asset-level provenance, builds four complementary indexes, understands and decomposes questions, plans an investigation, searches iteratively, expands through connected entities, verifies candidate claims, resolves conflicts, calculates confidence, and generates answers only from verified evidence. The user can see the final answer first, then inspect the evidence chain, source pages, graph path, search iterations, conflicts, confidence factors, and stopping reason.

The most important design rule is:

AI services may interpret, plan, rank, extract, and explain; they may never become the source of truth. The archive and its verified evidence objects remain the source of truth.

The challenge corpus contains mixed formats, approximately 415 documents and 1,277 pages, interconnected fictional entities, and sources with different reliability. The challenge explicitly states that public AI models have not seen this invented world, so the system must genuinely retrieve and reason over the supplied corpus.^[^1]

1. Problem Definition

Enterprise documentation is difficult to use because relevant information may be distributed across narrative text, tables, scanned pages, diagrams, images, maps, and records. A single search result may provide only one part of the answer. The user needs an assistant that can discover an initial fact, recognize what is missing, search again using what it learned, connect facts across documents, and clearly distinguish evidence from inference.

The official challenge defines Sub-track 1B around connecting facts across documents and Sub-track 1C around iterative human-like investigation: deciding where to look, checking whether enough information has been found, and continuing when evidence is incomplete.^[^1]

Selected scope

Capability	Implementation role
Sub-track 1C	Primary behavior: iterative planning, searching, evidence-gap detection, and stopping
Sub-track 1B	Cross-document entity linking and multi-hop graph reasoning
Sub-track 1A	Supporting capability: retrieve and display relevant tables, figures, diagrams, maps, and images

Product statement

A multimodal archive investigator that searches, connects, verifies, and explains information across an entire document collection with page-level evidence and an inspectable investigation trail.

2. Goals and Non-Goals

Goals

- Ingest PDFs, scanned PDFs, DOCX, Markdown, plain text, tables, images, figure plates, maps, diagrams, heraldry, and charts.
- Preserve document ID, file path, source type, page or line range, section, asset ID, OCR confidence, reliability, and original asset reference.
- Build keyword, vector, graph, and fact/claim indexes.
- Understand direct, multi-hop, comparison, timeline, visual, impact-analysis, and conflict-resolution questions.
- Plan and execute bounded iterative investigations.
- Connect facts distributed across documents and formats.
- Verify claims against original evidence.
- Detect and explain conflicting sources.
- Generate grounded answers with clickable citations.
- Display relevant tables, figures, and images.
- Continue working through deterministic fallback modes when AI APIs fail.
- Provide evaluation results, reproducible setup, and a credible live demonstration.

Non-goals

- The assistant is not a general web-search engine.
- It must not use external web knowledge to fill archive gaps.
- It must not present unsupported LLM suggestions as facts.
- It must not expose private hidden chain-of-thought.
- It must not silently modify the read-only source corpus.

- It must not claim complete certainty when the investigation stopped because of a budget or unresolved conflict.

3. Requirements

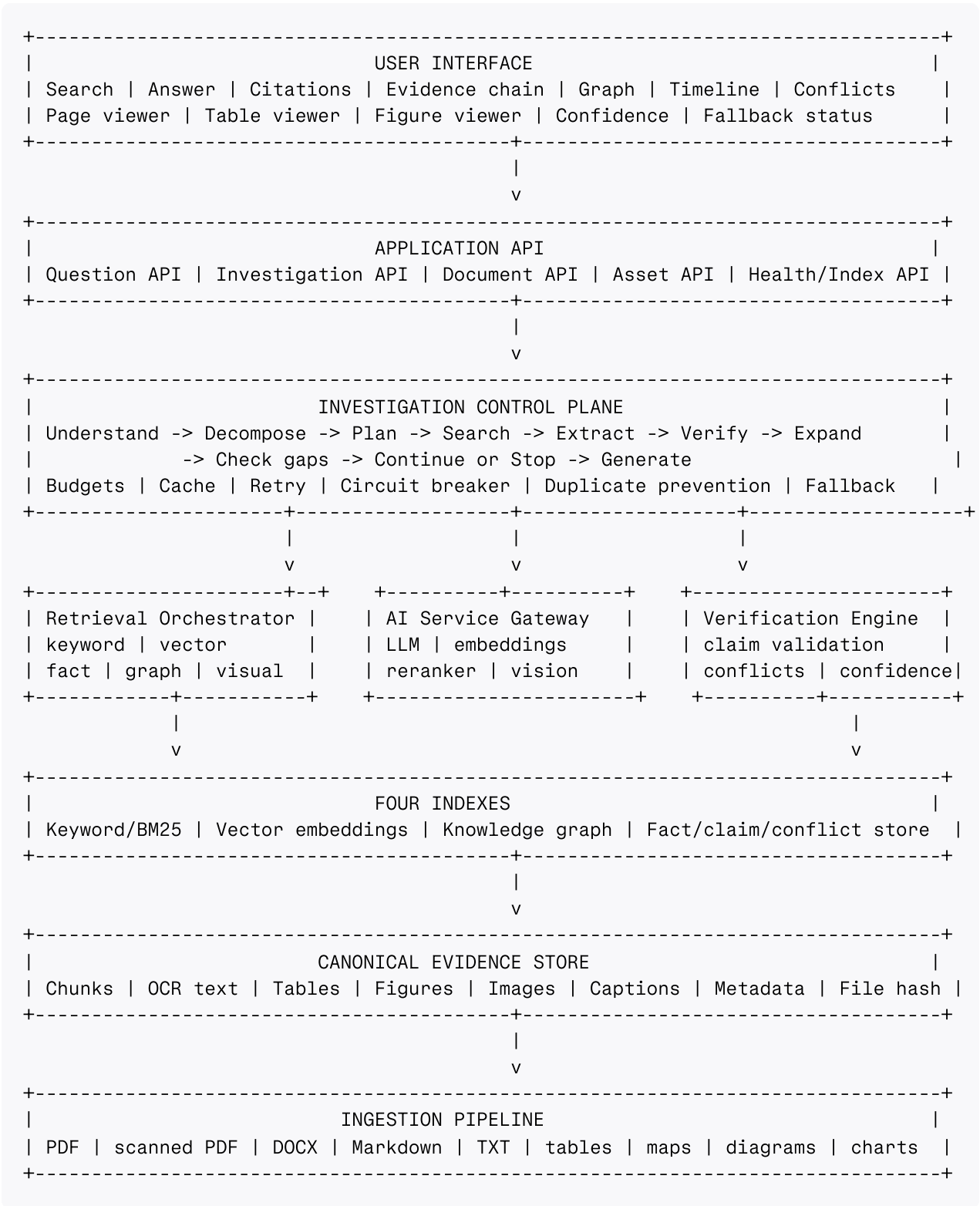
Functional requirements

ID	Requirement
FR-01	Ingest every supported archive format.
FR-02	Preserve complete provenance for every evidence object.
FR-03	Extract tables and visual assets as independent linked objects.
FR-04	Build and version four complementary knowledge indexes.
FR-05	Parse questions into validated structured representations.
FR-06	Decompose complex questions into sub-questions.
FR-07	Generate a bounded investigation plan.
FR-08	Search using exact, semantic, graph, fact, OCR, and visual routes.
FR-09	Extract and verify candidate entities, relations, and claims.
FR-10	Expand across graph-connected entities for multi-hop questions.
FR-11	Detect conflicts and apply source reliability rules.
FR-12	Generate answer, evidence chain, citations, confidence, uncertainty, and stop reason.
FR-13	Expose investigation trace, graph path, and linked assets in the UI.
FR-14	Provide deterministic fallbacks for external AI failures.
FR-15	Evaluate retrieval, reasoning, citations, conflicts, visuals, fallback behavior, cost, and latency.

Non-functional requirements

- Reproducible setup from a clean environment.
- Idempotent ingestion.
- Versioned indexes, prompts, and model configurations.
- Bounded latency, token usage, graph hops, and API calls.
- No API secrets in source code, logs, screenshots, or Git history.
- Auditable evidence and deterministic citation validation.
- Modular components that each team member can explain and modify.

4. High-Level Architecture



Architectural principles

- Provenance first:** no evidence object exists without source location.
- Hybrid retrieval:** no single index is trusted to answer every question.
- AI as a controlled tool:** every AI output is schema-validated and grounded.
- Bounded investigation:** every loop has explicit budgets and stopping conditions.

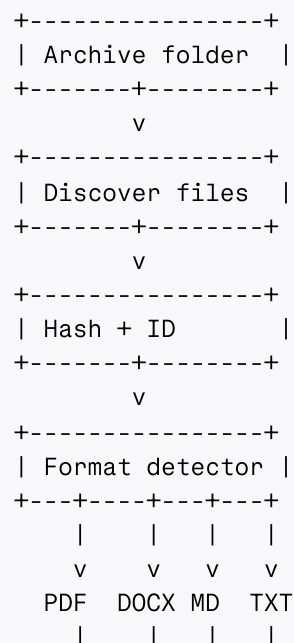
5. **Claim-level verification:** evidence is checked before answer generation.
6. **Visible uncertainty:** conflicts, OCR limitations, visual ambiguity, and partial results are shown.
7. **Failure tolerance:** the product degrades to deterministic archive search rather than hallucinating.

5. Archive Ingestion

Supported inputs

Input	Processing	Output
Selectable PDF	Text extraction while preserving pages and reading order	Page-aware text chunks
Scanned PDF	Page rendering, OCR, OCR confidence, original page image	OCR chunks plus page asset
DOCX	Paragraphs, headings, tables, headers, footers, embedded images	Section chunks, table objects, image objects
Markdown	Headings, paragraphs, tables, links, embedded images	Section-aware chunks and assets
Plain text	Line ranges and logical block segmentation	Line-aware chunks
Tables	Cell matrix, headers, captions, normalized text	Table object and structured facts
Images	File storage, metadata, OCR, caption and nearby-text linking	Image evidence object
Figure plates/maps/diagrams	OCR, caption, labels, nearby context, optional vision	Visual evidence object

Ingestion workflow



```

+----+---+---+-----> tables/images/OCR/assets
      |
      v
+-----+
| Structure normalizer |
| headings/pages/lines |
+-----+
      v
+-----+
| Canonical evidence   |
| chunks/tables/media  |
+-----+
      v
+-----+
| Build four indexes   |
+-----+

```

Canonical evidence record

```

{
  "evidence_id": "ev-doc17-p12-fig03",
  "document_id": "doc17",
  "document_version": 1,
  "original_path": "codex/beasts.pdf",
  "file_hash": "sha256:...",
  "source_type": "pdf_scan",
  "page_start": 12,
  "page_end": 12,
  "line_start": null,
  "line_end": null,
  "section_path": ["Codex of Beasts", "Gravemaw Wyrn"],
  "object_type": "figure",
  "table_id": null,
  "figure_id": "fig03",
  "caption": "Lair distribution map",
  "content": "OCR or normalized text representation",
  "nearby_evidence_ids": ["ev-doc17-p12-c01"],
  "ocr_confidence": 0.88,
  "vision_confidence": null,
  "source_reliability": "official_codex",
  "original_asset_ref": "assets/doc17/page12/fig03.png",
  "parser_version": "1.0.0"
}

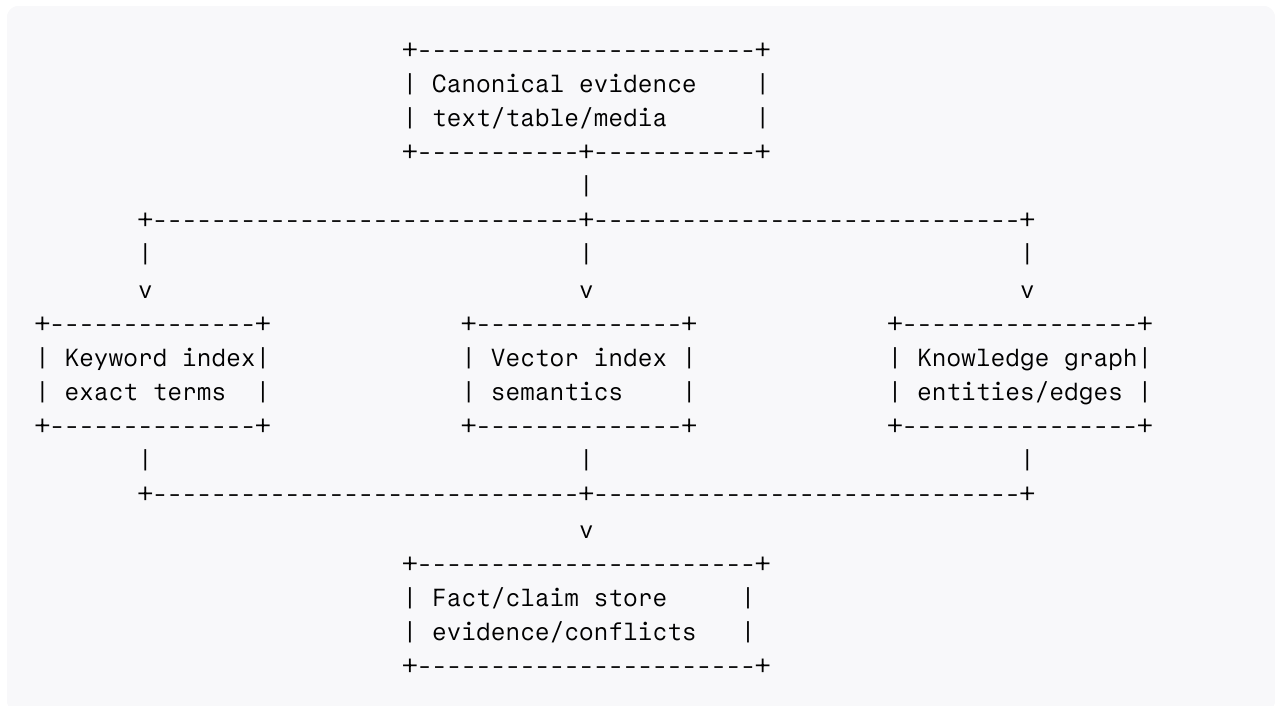
```

Provenance rules

- Every chunk points to a document, page or line range, and section.
- Every table stores its original page, caption, headers, rows, and nearby text.
- Every image or figure stores the original file reference, page, caption, and context.
- OCR and vision results are derived observations, not replacements for original assets.
- Evidence objects are immutable after creation; corrections create a new derived version.

- Ingestion is idempotent using file hashes and parser versions.
- The original corpus remains read-only.

6. Four Knowledge Indexes



Index responsibilities

Query need	Index route	Examples
Exact fictional name	Keyword/BM25	Character, faction, artifact, place
Date, number, title, alias	Keyword/BM25 + fact store	Dates and table values
Paraphrased question	Vector	Different wording and semantic similarity
Cross-document connection	Knowledge graph	lairs_in, ruled_by, depends_on
Structured fact	Fact store	Subject-predicate-object/value lookup
Conflicting claims	Claim/conflict store	Competing dates, owners, rulers
Visual question	Visual evidence + OCR + linked text	Maps, heraldry, diagrams, charts

Index versioning

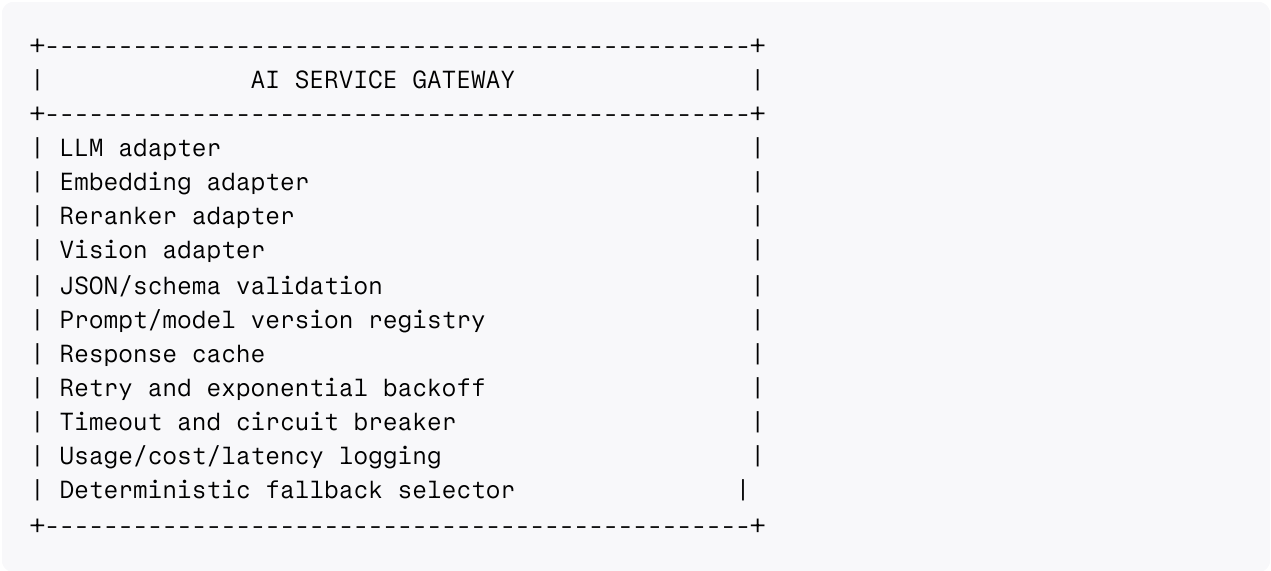
Each index stores:

- Corpus version.
- Document version set.
- Parser version.
- Embedding model and version.
- Graph extraction version.

- Fact extraction prompt version.
- Build timestamp.
- Configuration hash.

An investigation records the exact index versions used so that its result can be reproduced.

7. AI Service Gateway



Controlled AI responsibilities

Component	Permitted use	Restrictions
LLM	Question parsing, planning, query rewriting, candidate extraction, answer composition	Receives archive evidence; output must pass schema validation
Embedding model	Document and query vector representations	Model/version recorded; index versioned
Reranker	Ranking retrieved candidates	Can rank only archive-retrieved candidates
Vision model	Structured observations about diagrams, maps, charts, symbols, and images	Observation must retain figure ID, confidence, and support status

The final answer prompt receives only verified claims, valid citations, conflicts, confidence, relevant assets, and stop reason. It does not receive permission to fill gaps using general knowledge.

8. Question Understanding

Structured question object

```
{
  "question_id": "q-001",
  "raw_question": "Who rules the place where the Gravemaw Wyrmlives?",
  "intent": "multi_hop_relation_lookup",
  "main_topic": "Gravemaw Wyrmlives",
}
```

```

    "entities": [
      {"surface": "Gravemaw Wyrms", "canonical_id": null, "type": "creature"}
    ],
    "aliases": [],
    "relations": ["lairs_in", "ruled_by"],
    "answer_type": "ruler_or_faction",
    "requires_multi_hop": true,
    "requires_visual_check": false,
    "required_evidence": [
      "Wyrms location",
      "Location ruler",
      "Conflict check"
    ]
  }

```

Understanding process

1. Apply deterministic rules for question words, dates, numbers, quoted terms, and known aliases.
2. Use an LLM for intent, relation, answer type, and sub-question interpretation.
3. Validate JSON against a schema.
4. Match entities against the archive entity index.
5. Mark unknown entities as unresolved instead of inventing them.
6. Classify the question as direct lookup, multi-hop, comparison, timeline, visual, impact, or conflict resolution.

9. Investigation Planning

Planner output

The planner must define:

- Main question.
- Sub-questions.
- Initial entities and relations.
- Retrieval routes for each sub-question.
- Exact, semantic, alias, relation, and graph-neighbor query variants.
- Required evidence for completion.
- Whether tables, figures, OCR, or vision should be checked.
- Maximum iterations, graph hops, LLM calls, time, and query count.
- Expected stop conditions.

Example plan

Main question: Who rules the place where the Gravemaw Wyrms live?

Sub-question 1: Where is the Gravemaw Wyrms' lair?

Routes: fact lookup, exact search, vector search, graph neighbors

Completion: page-cited location claim

Sub-question 2: Who rules the discovered location?

Routes: exact location search, graph traversal, codex/wiki search

Completion: page-cited ruler claim

Sub-question 3: Do authoritative sources disagree?

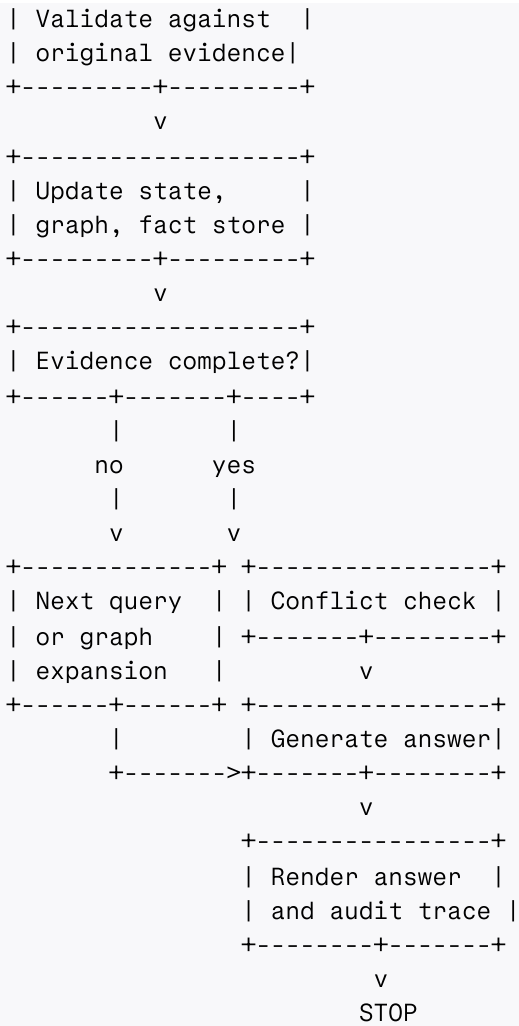
Routes: claim store conflict lookup, source-type comparison

Completion: conflict checked and classified

Global completion: all required evidence conditions satisfied

10. Iterative Investigation Loop

```
+-----+
| START |
+---+---+
      v
+-----+
| Understand query |
+-----+
      v
+-----+
| Decompose question|
+-----+
      v
+-----+
| Create plan      |
+-----+
      v
+-----+
| Search current gap|
| fact/keyword/    |
| vector/graph/    |
| OCR/visual       |
+-----+
      v
+-----+
| Merge/deduplicate |
| and rerank        |
+-----+
      v
+-----+
| Extract candidates|
| entities/claims   |
+-----+
      v
+-----+
```



Search routes

For each active evidence gap, the system may use:

- Exact fact lookup.
- Keyword/BM25 search.
- Vector semantic search.
- Knowledge graph traversal.
- Linked tables and figure retrieval.
- OCR search.
- Vision analysis when text and OCR are insufficient.

The system merges and deduplicates candidates, reranks them, reads the strongest evidence, extracts candidate facts, validates each fact against the original evidence, and updates the investigation state.

Iteration record

```
{
  "iteration": 2,
  "active_sub_question": "Who rules Marrowwell Abbey?",
  "queries": [
    "Marrowwell Abbey ruler",
    "Marrowwell Abbey ruled by",
    "Marrowwell Abbey faction"
  ],
  "routes": ["keyword", "vector", "graph", "fact_store"],
  "retrieved_evidence": ["ev-doc17-p42-c03", "ev-doc88-p11-c02"],
  "new_entities": ["Bleeding Crown"],
  "candidate_claims": ["Marrowwell Abbey is ruled by the Bleeding Crown"],
  "verified_claims": ["claim-042"],
  "remaining_gaps": ["conflict check"],
  "next_action": "query conflict store",
  "decision_reason": "ruler evidence found; conflict requirement remains"
}
```

Stopping policy

Stop when:

1. Every required sub-question has valid supporting evidence.
2. No useful new evidence appears after a configured number of iterations.
3. The investigation budget is reached.
4. The user requests a shorter response.

The interface must state the actual stop reason. Budget and no-progress stops produce partial or low-confidence output when completion conditions remain unsatisfied.

11. Claims, Facts, and Verification

Claim record

```
{
  "claim_id": "claim-042",
  "subject_id": "place-marrowwell-abbey",
  "predicate": "ruled_by",
  "object_id": "faction-bleeding-crown",
  "claim_type": "direct",
  "source_evidence_ids": ["ev-doc42-p34-c01"],
  "evidence_text_span": "...",
  "page": 34,
  "extraction_method": "structured_llm_extraction",
  "source_reliability": "official_codex",
  "ocr_confidence": null,
  "vision_confidence": null,
  "claim_confidence": 0.94,
}
```

```
"verification_status": "verified"
}
```

Claim classifications

Class	Definition	Output rule
Direct	Explicitly stated by a source	May be presented directly
Inferred	Derived from a documented chain of verified claims	Show all supporting links
Conflicting	Competing values supported by different evidence	Report disagreement and reliability
Unsupported	Suggested but not proven by archive evidence	Exclude from factual answer
Partial visual	Observation has incomplete confirmation	Show image and uncertainty

Verification gate

A claim is verified only if:

- Its evidence object exists.
- The page or line range is valid.
- The evidence supports the subject, predicate, and object/value.
- The claim was not accepted solely because an LLM generated it.
- OCR or vision limitations are recorded.
- Source reliability is known.

12. Source Reliability and Conflict Resolution

The archive contains sources with different levels of reliability, and the challenge explicitly notes that official codex entries may disagree with less authoritative material such as a tavern ballad.^[^1]

Reliability policy

Source type	Default reliability	Interpretation
Official codex/data record	Highest	Strongest for structured facts and specifications
Primary official document	High	Strong for recorded events and decisions
Wiki article	Medium	Useful synthesis; verify important claims
Novel narrative	Contextual	Strong for narrated events, but viewpoint may be limited
Letter/ledger/trial transcript	Context-dependent	Assess author, purpose, and proximity to event
Ballad/rumor	Low	Use as lead or conflict evidence unless corroborated

Conflict detection

Normalize claims into subject–predicate–object/value triples. Group claims with the same subject and predicate. If values differ, create a conflict record containing:

- Competing claims.
- Supporting evidence.
- Page and section references.
- Source categories.
- Reliability scores.
- Whether one claim is direct and another is contested or inferred.
- Whether the conflict can be resolved.

Conflict response example

The Codex records Gloamreach as founded in 246 AS. A wiki article marks that date as contested. Because the codex is the more authoritative source, 246 AS is the best-supported recorded date, while the secondary conflict is reported rather than hidden.

13. Confidence Model

Confidence is calculated from explicit factors rather than generated as an unexplained model opinion:

- Source reliability.
- Number of independent supporting documents.
- Direct versus inferred status.
- Graph-hop count.
- OCR confidence.
- Vision-analysis confidence.
- Evidence quality.
- Citation validity.
- Unresolved conflict penalty.
- Unsupported or missing sub-question penalty.

Store each factor separately. Present a human-readable confidence label such as **High**, **Medium**, **Low**, or **Partial visual**, together with the reasons.

14. Visual Evidence Handling

Visual pipeline

```
Relevant figure/image retrieved
|
v
Original asset + caption + nearby text
|
+--> OCR labels and text
|
+--> Vision model if necessary
|
v
Structured observation + confidence
|
v
Evidence verification
|
+--> confirmed by labels/context
+--> partially supported
+--> unresolved visual observation
|
v
Answer with original asset and uncertainty
```

Visual evidence rules

- Always preserve and link the original image or page asset.
- Read the caption and nearby text before invoking vision analysis.
- Use OCR for labels, numbers, titles, and visible text.
- Invoke a vision-capable model only when OCR and context are insufficient.
- Store structured visual observations with figure ID and model confidence.
- Require supporting labels, captions, nearby text, or repeated evidence when possible.
- Never invent unreadable visual facts.

Safe responses

If the visual evidence cannot be confirmed:

The answer is not established in readable text. The relevant evidence appears to be pictorial on the identified figure plate. The original image is available for manual review.

If partially supported:

A vision model identifies a central black stag-like emblem, but the plate contains no readable confirming label. This is partial visual evidence rather than a confirmed archive fact.

15. Grounded Answer Generation

Generator input contract

The final answer generator receives only:

- Verified direct claims.
- Verified inferred claim chains.
- Valid citations.
- Conflict records.
- Confidence factors.
- Relevant tables, figures, and images.
- Investigation trace summary.
- Stop reason.

Answer format

FINAL ANSWER

Concise answer to the question.

EVIDENCE CHAIN

1. Direct fact with document and page citation.
2. Direct fact with document and page citation.
3. Inferred conclusion showing how the facts connect.

CLAIM TYPE

Direct / Inferred / Conflicting / Partial visual

CONFIDENCE

High / Medium / Low, including reasons.

CONFLICTS AND UNCERTAINTY

Competing sources, OCR limitations, visual ambiguity, or missing evidence.

RELATED EVIDENCE

Clickable document pages, tables, figures, and image assets.

INVESTIGATION TRACE

Queries, retrieval routes, graph expansions, and stop reason.

Example

The Bleeding Crown rules the territory where the Gravemaw Wyrms lairs.

Evidence chain:

1. The Gravemaw Wyrms' lair is identified as Marrowwell Abbey [Codex of Beasts, p. 12].
2. Marrowwell Abbey is recorded as ruled by the Bleeding Crown [Regional Codex, p. 34].

Claim type: Inferred from two direct claims.

Confidence: High.

Stop reason: All required sub-questions have page-cited evidence and no unresolved authoritative conflict was found.

16. API Design

Question and investigation APIs

```
POST /api/questions
GET  /api/investigations/{investigation_id}
GET  /api/investigations/{investigation_id}/trace
GET  /api/investigations/{investigation_id}/evidence
GET  /api/investigations/{investigation_id}/graph
```

Archive and asset APIs

```
GET  /api/documents/{document_id}
GET  /api/documents/{document_id}/pages/{page}
GET  /api/assets/{asset_id}
GET  /api/indexes/status
GET  /api/health
```

Example request

```
{
  "question": "Who rules the place where the Gravemaw Wyrms lives?",
  "options": {
    "max_iterations": 6,
    "max_graph_hops": 3,
    "max_llm_calls": 12,
    "include_visual_evidence": true,
    "show_trace": true,
    "fallback_allowed": true
  }
}
```

Example response

```
{
  "investigation_id": "inv-2026-0001",
  "status": "completed",
  "answer": "The Bleeding Crown rules ...",
  "claims": ["claim-041", "claim-042", "claim-043"],
  "evidence_ids": ["ev-doc17-p12-c02", "ev-doc42-p34-c01"],
  "graph_paths": [
    ["Gravemaw Wyrms", "lairs_in", "Marrowwell Abbey", "ruled_by", "Bleeding Crown"]
  ],
  "conflicts": [],
  "confidence": "high",
  "stop_reason": "required_evidence_supported",
  "ai_mode": "llm_assisted"
}
```

17. User Interface

```
+-----+
|                                     |
|               ASHEN ARCHIVE INTELLIGENCE               |
|                                     |
+-----+
| [ Ask a question about the archive... ] [Ask] |
+-----+
| FINAL ANSWER                               Confidence: High |
| The Bleeding Crown rules the territory ... |
| Claim type: Inferred |
+-----+
| EVIDENCE CHAIN |
| 1. Wyrms -> lairs_in -> Abbey [Open document p.12] |
| 2. Abbey -> ruled_by -> Bleeding Crown [Open document p.34] |
+-----+
| CLAIM AND SOURCE DETAILS |
| Direct | Inferred | Conflicting | Source reliability | Citation validity |
+-----+
| RELATED VISUAL/TABLE EVIDENCE |
| [table preview] [figure preview] [Open original asset] |
+-----+
| INVESTIGATION TIMELINE |
| Understand -> Plan -> Search 1 -> Graph expansion -> Search 2 -> Verify -> Stop |
+-----+
| KNOWLEDGE GRAPH |
| Wyrms ----lairs_in----> Abbey ----ruled_by----> Bleeding Crown |
+-----+
| CONFLICTS / UNCERTAINTY |
| None found, or competing claims with source explanation |
+-----+
```

Required UI features

- Search box.
- Final answer first.
- Clickable page-level citations.
- Evidence chain.
- Direct, inferred, conflicting, and partial-visual labels.
- Confidence and confidence factors.
- Source reliability.
- Conflict panel.
- Related documents.
- Openable pages, tables, figures, and images.
- Investigation timeline.
- Search queries performed.
- Knowledge graph path.
- Stop reason.
- LLM-assisted or deterministic-fallback status.

18. Reliability, Security, and Fallbacks

Security controls

- Store API keys only in environment variables or ignored `.env` files.
- Never commit keys to Git.
- Never include secrets in logs, screenshots, chat histories, or error messages.
- Validate document IDs and asset paths to prevent path traversal.
- Restrict asset access to known archive objects.
- Redact provider and configuration secrets from logs.
- Rotate leaked keys immediately.

The competition guidance specifically warns that exposed keys remain visible in old Git commits and recommends environment variables, caching, and exponential backoff for rate limits.^[^1]

Operational controls

- Strict JSON schema validation.
- Prompt, model, and index version tracking.
- Embedding cache.
- LLM plan and extraction cache.

- Reranker and vision cache.
- Completed-investigation cache.
- Retry with exponential backoff.
- Timeouts and circuit breakers.
- Maximum iteration, graph-hop, token, cost, and latency budgets.
- Duplicate query detection.
- Citation existence and support validation.
- Deterministic fallback mode.

Fallback matrix

Failure	Fallback	Output mode
LLM unavailable	Rules plus deterministic planner	deterministic_fallback
Embedding API unavailable	BM25/FTS5, fact store, graph	keyword_graph_mode
Vision unavailable	OCR, captions, nearby text	visual_analysis_unavailable
Reranker unavailable	Weighted transparent ranking	default_ranking
Graph unavailable	Fact store plus text retrieval	graph_unavailable
Rate limit	Cache, retry, alternate provider, or reduced plan	provider_fallback

The fallback system must reduce capability transparently, never replace missing evidence with hallucinated content.

19. Evaluation Strategy

The archive provides a 20-question development set, while final judging uses a separate unpublished set of similar style. Evaluation must therefore measure general capabilities rather than hard-coded answers.^[1]

Ablation modes

1. Keyword-only.
2. Vector-only.
3. Keyword plus vector.
4. Hybrid plus graph expansion.
5. Deterministic investigation loop.
6. LLM-assisted investigation loop.
7. Full multimodal system.

Metrics

Metric	Measurement
Cited-answer coverage	Percentage of factual answer claims with citations
Citation correctness	Whether each citation actually supports the claim
Relevant page recall	Relevant pages retrieved for each question
Semantic retrieval success	Paraphrased question performance
Multi-hop completion	Questions requiring connected facts
Conflict detection	Competing source identification
Unsupported claim count	Ungrounded factual claims in output
Visual evidence accuracy	Correct visual observations and uncertainty handling
OCR robustness	Performance on scans and OCR errors
Fallback success	Useful answers when APIs fail
Latency	Time to answer
Cost per question	External model usage after caching
Reproducibility	Stability under repeated runs

Test categories

- Unit tests for parsers, provenance, schemas, scoring, and citation validation.
- Integration tests for ingestion-to-index and question-to-answer flows.
- Multi-hop tests with three or more graph edges.
- Table and figure retrieval tests.
- OCR corruption tests.
- Conflicting-source tests.
- No-answer and insufficient-evidence tests.
- API timeout and rate-limit tests.
- Deterministic fallback tests.
- Security tests for secret leakage and path traversal.

20. Technology Stack

Layer	Recommended	Lightweight alternative
Backend	FastAPI	Spring Boot API gateway or Flask
PDF	PyMuPDF	pdfplumber where suitable
DOCX	python-docx	LibreOffice conversion plus parser

Layer	Recommended	Lightweight alternative
OCR	Tesseract or configured OCR API	Local OCR engine
Keyword	OpenSearch/Elasticsearch	SQLite FTS5
Vector	Qdrant/pgvector	Chroma local mode
Graph	Neo4j/Memgraph	NetworkX or relational edges
Fact store	PostgreSQL	SQLite
Assets	S3-compatible storage	Local filesystem
Frontend	React/Next.js	Minimal React UI
AI access	OpenAI-compatible gateway	Provider adapters behind common interface
Packaging	Docker Compose	Virtual environment and scripts
Testing	Pytest and evaluation runner	Scripted JSON test harness

The challenge permits any programming language, framework, model, or service and states that paid services are not required. The lightweight alternative preserves the same logical architecture while reducing deployment risk.[^1]

21. Repository Structure

```

<team-name>/
├── .git/
├── README.md
├── pyproject.toml
├── docker-compose.yml
├── .env.example
├── src/
│   ├── api/
│   ├── config/
│   ├── ingestion/
│   ├── evidence/
│   ├── indexes/
│   │   ├── keyword/
│   │   ├── vector/
│   │   ├── graph/
│   │   └── claims/
│   ├── ai_gateway/
│   ├── investigation/
│   ├── verification/
│   ├── generation/
│   ├── fallback/
│   └── common/
├── tests/
│   ├── unit/
│   ├── integration/
│   ├── evaluation/
│   └── failure_modes/
├── scripts/
└── ingest_archive.py

```

```

|   ├── build_indexes.py
|   ├── reindex.py
|   └── run_evaluation.py
|   └── docs/
|       ├── architecture.md
|       ├── data-model.md
|       ├── investigation-protocol.md
|       ├── api.md
|       ├── decisions.md
|       ├── limitations.md
|       └── diagrams/
|   └── data/
|       └── README.md
|   └── ai_usage/
|       ├── ai-usage-disclosure.md
|       └── chat-logs/
|   └── configuration-example/
|   └── submission_report.pdf

```

The official submission requires a complete Git repository with history, README, architecture and design documentation, diagrams, limitations, AI usage disclosure, exported AI chat logs, and a maximum five-page PDF report.^[1]

22. Team Responsibilities

Role	Responsibilities	Judging explanation
Ingestion/provenance engineer	Format parsers, OCR, tables, visual assets, stable IDs	Explain page fidelity and provenance
Retrieval/index engineer	BM25, vector, graph, fact store, fusion	Explain retrieval routes and ranking
AI/investigation engineer	Gateway, planner, loop, budgets, fallbacks	Explain controlled AI and stopping
Verification/product engineer	Claims, conflicts, UI, evaluation, demo	Explain grounding and measurable quality

All four members should be able to explain the complete system. The challenge states that team members may be asked to explain and modify any part of the solution, and that generic one-shot AI output is not sufficient.^[1]

23. Key Technical Decisions

Hybrid retrieval

Keyword search is precise for fictional names, aliases, numbers, and table values. Vector search handles paraphrases. The graph connects entities and relations. The fact store handles structured claims and conflicts. Combining them is more robust than relying on any single method.

Four indexes instead of one database

Separating retrieval responsibilities makes failures diagnosable and enables ablation testing. It also prevents a graph edge, vector match, or LLM statement from being mistaken for a verified fact without provenance.

Bounded state machine

A state machine provides human-like iterative investigation while remaining reproducible, debuggable, rate-limit aware, and suitable for a live demo. Every loop has a reason, an evidence gap, a budget, and a stop condition.

AI gateway

A single gateway enables provider switching, caching, retries, circuit breakers, usage logging, and deterministic fallback behavior. It also makes external AI use easier to disclose and explain.

Claim verification before generation

The language model should make evidence understandable, not decide whether evidence is true. Verification occurs before final generation, and unsupported claims are excluded.

Visual uncertainty

Visual evidence is valuable but ambiguous. Storing original assets, OCR, captions, visual observations, and confidence allows the system to be useful without pretending that unreadable images are certain facts.

24. Limitations and Mitigations

Limitation	Risk	Mitigation
OCR errors	Names, dates, and table values may be wrong	OCR confidence, aliases, fuzzy search, original page viewer
Vision ambiguity	Symbols or layouts may be misinterpreted	Require labels/context, show uncertainty, preserve original asset
Entity resolution	Similar entities may be merged	Canonical IDs, alias records, confidence and trace
Incomplete graph	Valid relationships may be missing	Search graph and source indexes together
LLM extraction errors	False entities or claims	Schema validation and evidence verification
Conflicting sources	Multiple archive narratives	Report conflict, reliability, and unresolved status
API rate limits	Interrupted investigation	Cache, retry, backoff, provider fallback
Large graph neighborhoods	Slow or noisy investigation	Relation filters, hop limits, deduplication
Unpublished evaluation questions	Overfitting	General capability tests and no question-specific rules

Limitation	Risk	Mitigation
Large visual assets	Slow UI	Thumbnails, lazy loading, original on demand

25. Demonstration Plan

The competition requires a video no longer than ten minutes, with at least three to four minutes of live, unedited screen recording demonstrating the working prototype on real inputs.^[^1]

Recommended live demonstration

1. Show archive ingestion completion.
2. Show four index health status.
3. Ask a multi-hop question.
4. Display structured question understanding.
5. Display the initial search routes.
6. Show evidence retrieved from different documents.
7. Show graph expansion to a newly discovered entity.
8. Show the next query generated from the remaining evidence gap.
9. Show claim validation and source reliability.
10. Demonstrate conflict handling or visual evidence.
11. Present the final answer with clickable page citations.
12. Open a cited page, table, or image.
13. Show the investigation timeline and stop reason.
14. Briefly demonstrate deterministic fallback or an API failure simulation.

The live segment should demonstrate genuine system behavior rather than only a prewritten response.

26. Implementation Roadmap

Phase 1: Vertical slice

Implement selectable-text PDF ingestion, page-aware chunks, keyword search, one vector route, a question endpoint, citations, and a basic answer view.

Phase 2: Complete ingestion

Add scans, OCR, DOCX, Markdown, plain text, tables, images, figures, maps, captions, and asset linking.

Phase 3: Four-index fabric

Add vector index, graph, entity resolution, fact store, claim schema, conflict grouping, and index health.

Phase 4: Investigation controller

Add structured question understanding, sub-question planning, iterative search, graph expansion, evidence-gap tracking, budgets, and stop reasons.

Phase 5: Verification and multimodal support

Add claim validation, reliability, confidence factors, contradiction detection, OCR uncertainty, visual analysis, and safe partial answers.

Phase 6: Product hardening

Add UI trace, clickable citations, page and asset viewer, caching, retries, circuit breakers, fallback modes, tests, and Docker setup.

Phase 7: Submission readiness

Complete README, architecture diagrams, decisions, limitations, evaluation results, AI usage disclosure, exported chat logs, meaningful commits, demonstration video, and five-page PDF report.

27. Submission Checklist

Working system

- [] PDF, scanned PDF, DOCX, Markdown, TXT, tables, images, figures, maps, diagrams, and charts are handled.
- [] Every evidence object has complete provenance.
- [] Four indexes build successfully.
- [] Query understanding and planner produce validated JSON.
- [] Iterative investigation loop works.
- [] Graph expansion handles multi-hop questions.
- [] Claims are verified before answer generation.
- [] Conflicts and source reliability are visible.
- [] Visual evidence includes uncertainty handling.
- [] Fallback modes work when AI services fail.
- [] Citations open the correct page or asset.

Engineering and documentation

- ☐ README works from a fresh environment.
- ☐ Dependencies are pinned.
- ☐ `.env.example` is included and secrets are excluded.
- ☐ Unit, integration, evaluation, and failure-mode tests exist.
- ☐ Architecture, data model, API, decisions, and limitations are documented.
- ☐ Git history contains meaningful atomic commits.
- ☐ AI usage disclosure is included.
- ☐ AI chat logs are exported as required.

Competition deliverables

- ☐ Demonstration video is ten minutes or less.
- ☐ Video includes at least three to four minutes of live, unedited demonstration.[^1]
- ☐ YouTube video is uploaded as unlisted and linked in the report.[^1]
- ☐ Full Git repository ZIP includes history.[^1]
- ☐ PDF report is five pages or less and includes problem, track, architecture, decisions, limitations, AI disclosure, and team contributions.[^1]

28. Final Positioning

The final solution should be presented as:

A provenance-first multimodal archive investigator that combines four knowledge indexes with bounded iterative planning, multi-hop graph reasoning, claim verification, conflict-aware confidence, visual evidence handling, and reliable fallback behavior.

Its competitive advantage is the complete evidence lifecycle: preserve the original archive, retrieve through complementary indexes, investigate until evidence requirements are satisfied, validate every claim, expose conflicts and uncertainty, and show the user exactly what was searched and why the system stopped. This directly addresses the mixed formats, cross-document relationships, source disagreements, and iterative-search requirements of the Ashen Era Archive challenge.[^1]

References

1. [AI Competition Challenge Final.pdf](#)